

 Part 1: What is NLP?

◆ **Definition**

NLP (Natural Language Processing) is a branch of Artificial Intelligence that enables computers to understand, process, analyze, and generate human language.

Human Language Examples:

- English
- Malayalam
- Hindi
- Tamil
- Arabic

NLP helps computers understand these languages.

Example

When you type:

What is AI?

ChatGPT understands your question and generates an answer.

This is NLP in action.

 Part 2: Why Do We Need NLP?

Computers understand:

0 and 1

(Binary Data)

Humans communicate using:

Words

Sentences

Language

NLP acts as a bridge between humans and computers.

Visual Representation

Human Language



NLP



Computer Understanding

Part 3: Real-World Applications of NLP

Chatbots

Examples:

- ChatGPT
 - Customer Support Bots
-

Voice Assistants

Examples:

- Google Assistant
 - Siri
 - Alexa
-

Language Translation

Example:

English → Malayalam

Malayalam → English

Spam Detection

Email systems detect:

Spam
Not Spam

Sentiment Analysis

Detect:

Positive Review

Negative Review

Neutral Review

Part 4: What is Text Data?

NLP mainly works with text data.

Examples:

Hello

How are you?

AI is amazing.

Types of Text Data

Structured Text

Database Records

Unstructured Text

Emails

Messages

Social Media Posts

Reviews

Most NLP works on unstructured text.

Part 5: NLP Workflow

Text Data
↓
Text Cleaning
↓
Text Processing
↓
Feature Extraction
↓
Model Training
↓
Prediction

Part 6: NLP Examples

Example 1

Input:

I love AI

Output:

Positive Sentiment

Example 2

Input:

This product is bad

Output:

Negative Sentiment

Example 3

Input:

Translate Hello to Malayalam

Output:

Part 7: Challenges in NLP

Human language is complex.

Same Word – Different Meaning

Example:

Bank

Meaning 1:

Financial Institution

Meaning 2:

River Bank

Computers must understand context.

Different Languages

Examples:

English

Malayalam

Hindi

Tamil

Each language has different grammar.

Part 8: Common NLP Tasks

Text Classification

Example:

Spam

Not Spam

Sentiment Analysis

Example:

Positive

Negative

Neutral

Translation

Example:

English → Malayalam

Question Answering

Example:

ChatGPT

Chatbot Development

Example:

Customer Support Bot

Part 9: NLP vs Computer Vision

NLP

Works with Text

Language Processing

Chatbots

Computer Vision

Works with Images

Image Processing

Object Detection

NLP

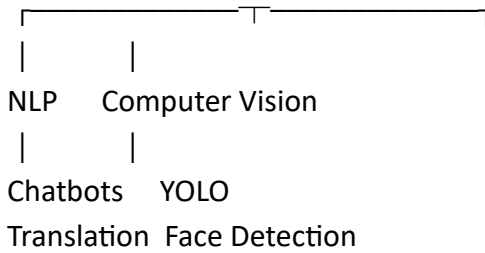
Translation

Computer Vision

Face Detection

📌 Part 10: AI Domains Overview

Artificial Intelligence



📌 Part 11: Examples Around Us

WhatsApp

Predictive typing.

Gmail

Spam detection.

YouTube

Comment analysis.

Amazon

Review analysis.

ChatGPT

Conversation generation.

📌 Part 12: Importance of NLP in AI

Without NLP:

- ✗ No ChatGPT
 - ✗ No Voice Assistant
 - ✗ No Translation Apps
 - ✗ No Chatbots
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With NLP:

- ✓ Smart Assistants
 - ✓ Chatbots
 - ✓ Translation Systems
 - ✓ Text Analysis
-

📌 Part 13: Theory Questions

What is NLP?

Natural Language Processing is a branch of AI that helps computers understand human language.

Why is NLP important?

It enables communication between humans and computers.

Give three applications of NLP.

1. Chatbots
 2. Translation
 3. Sentiment Analysis
-

What is Text Data?

Data represented as words, sentences, and language.

Difference between NLP and Computer Vision?

NLP works with text, while Computer Vision works with images and videos.

Part 14: Interview Questions

What is NLP?

NLP enables computers to understand human language.

What are the applications of NLP?

- Chatbots
 - Translation
 - Sentiment Analysis
 - Spam Detection
-

Why is NLP difficult?

Because human language is complex and context-dependent.

What is Sentiment Analysis?

Determining whether text is positive, negative, or neutral.

What is a Chatbot?

A software application that communicates with users through text or voice.

Part 15: Real-World NLP Projects

Chatbot

College Enquiry Bot

Review Analyzer

Amazon Review Sentiment Analysis

Translator

English ↔ Malayalam Translator

Email Spam Detector

Classify emails as spam or not spam.

FAQ Bot

Answer common questions automatically.

Tasks for Day 51

Theory

1. What is NLP?
 2. Why do we need NLP?
 3. Explain three NLP applications.
 4. Difference between NLP and Computer Vision.
 5. What is text data?
-

Practical Task 1

List 10 NLP applications used in daily life.

Practical Task 2

Identify 5 apps that use NLP.

Practical Task 3

Collect 20 positive and 20 negative reviews from the internet.

Practical Task 4

Write a short paragraph explaining how ChatGPT uses NLP.

Practical Task 5

Draw the NLP workflow diagram.